

A precise causal-set question for David A. Meyer

Does a vanishing-mass planted chain give a genuine $F1 + F2 \not\Rightarrow F3$ counterexample?

Ignacio Martín Gandul · nachocausal research programme · WP7 · 11 August 2026

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WP7

Under Madsen’s regional reading, does the fixed- d family $P_\rho = \Pi_\rho \cup \Gamma_\rho$, now proved for every $d \geq 2$, count as a faithful, count-uniform counterexample to $F1 + F2 \Rightarrow F3$ — or is there an intended causal-set condition that excludes it?

MATHEMATICS CLOSED UNDER STATED READING

EXPERT ADJUDICATION REQUESTED

PROVED REGIME: EVERY FIXED $d \geq 2$

We are not asking whether the two estimates can be proved: they can. We are asking whether the construction matches the intended causal-set semantics and whether the same mechanism is already present in the literature.

What Madsen separates

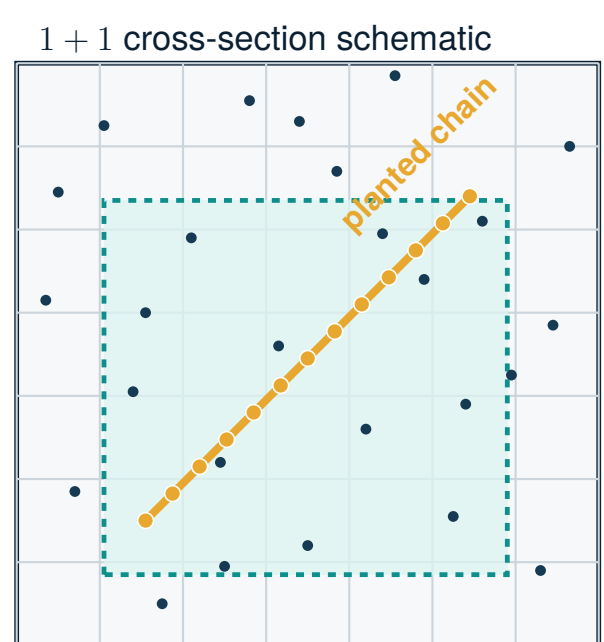
For a well-conditioned causal-set embedding:

- F1** Exact preservation of the induced causal order.
- F2** Uniform number–volume control in every admissible mesoscopic diamond.
- F3** Longest-chain distance approximates Lorentzian proper time at the prescribed rate.

With $\mathcal{N}_\rho = \rho V_K$, the relevant F2 tolerance is

$$|\#(P_\rho \cap D) - \rho \text{Vol}_g(D)| \leq K_2 \sqrt{\rho \text{Vol}_g(D) \log \mathcal{N}_\rho}.$$

The proposed family



Start with a background $\Pi_\rho \subset K$ satisfying F1–F2. Along a fixed timelike geodesic segment of length τ_0 , add

$$\Gamma_\rho, \quad k_\rho = \left\lceil A(m_d \rho)^{1/d} \tau_0 \right\rceil.$$

In every fixed dimension:

- ▶ $\gamma_0 \cap D$ is an interval;
- ▶ $\text{len}(\gamma_0 \cap D) \leq \tau(D)$;
- ▶ $\text{Vol}(D) \geq \kappa_d \tau(D)^d$.

The added mass fraction is only $\Theta(\rho^{1/d-1})$, but it is perfectly aligned in the order.

Geometric realization used for the check:

$$M_\ell^d = (-\infty, 0)_\eta \times \mathbb{R}_x^{d-1}, \quad g = \frac{\ell^2}{\eta^2}(-d\eta^2 + d\mathbf{x}^2)$$

on a precompact regional domain with a deep witness diamond.

Dimension: theorem for every fixed $d \geq 2$

In d -dimensional planar de Sitter, slice integration gives a fixed $\kappa_{d,\beta} > 0$ such that

$$\text{len}(\gamma_0 \cap D) \leq \tau(D) \leq \kappa_{d,\beta}^{-1/d} \text{Vol}(D)^{1/d}.$$

At Madsen’s minimum scale, the normalized planted cost is

$$O\left((\log \mathcal{N}_\rho)^{2/d-3/2}\right) \rightarrow 0.$$

Thus the rates are $-1/2$ in $1+1$, $-5/6$ in $2+1$, and -1 in $3+1$. The geometric transfer and exact F3 normalization are proved, not conjectured.

Estimate 1: the planted chain is invisible to F2

Causal convexity makes $\gamma_0 \cap D$ an interval, and a uniform volume–trace comparison gives

$$\#(\Gamma_\rho \cap D) \leq A m_d^{1/d} \kappa_{d,\beta}^{-1/d} (\rho \text{Vol}_g(D))^{1/d} + 2.$$

With $B_d := A m_d^{1/d} \kappa_{d,\beta}^{-1/d}$, the additional F2 cost is

$$\Xi_{A,d}(\rho) = B_d (\kappa_{d,\beta} c_*^{-d})^{1/d-1/2} (\log \mathcal{N}_\rho)^{2/d-3/2} + O((\log \mathcal{N}_\rho)^{-3/2}) \rightarrow 0.$$

Hence the final configuration still satisfies F2 with a fixed constant K_d .

Estimate 2: the same chain is visible to F3

For the planted endpoints $x_\rho \prec y_\rho$,

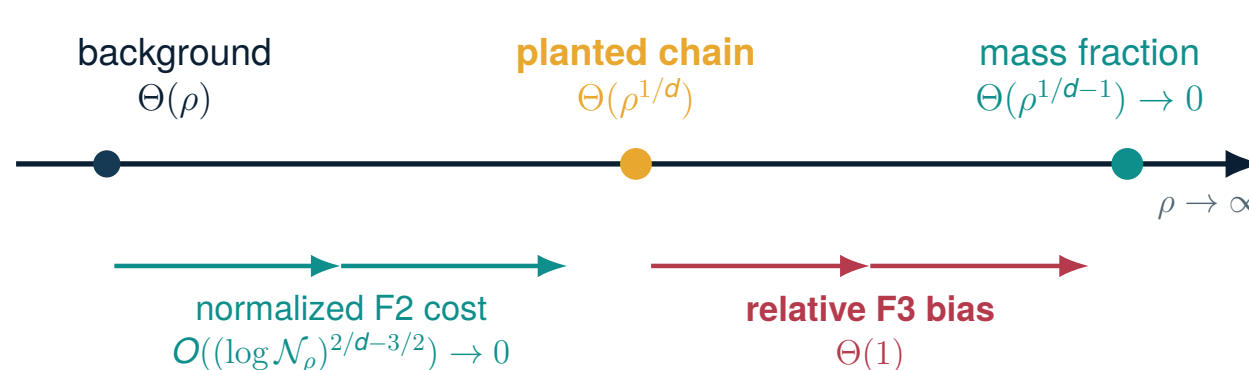
$$\frac{\ell_{P_\rho}(x_\rho, y_\rho)}{(m_d \rho)^{1/d}} - \tau_0 \geq (A-1)\tau_0 - o(1).$$

Madsen’s F3 bound would allow only

$$C_d \left(\varepsilon^2 + \frac{\log^{3/2} \mathcal{N}_\rho}{\mathcal{N}_\rho^{1/(2d)}} \right) \tau_0.$$

Choose $A > 1 + C_d \varepsilon^2$. Then $\mu = A - 1 - C_d \varepsilon^2 > 0$, and for large ρ the full F3 inequality fails by at least $\mu \tau_0 / 3$.

Why both statements coexist



The perturbation is asymptotically negligible for counts but macroscopically visible to chain height.

O5: PROVED_MODULO_O3_SCOPE

P5.2: PASS_WITH_SCOPE

What is already closed mathematically

- ▶ $d = 2$: deterministic product-order construction and full geometric transfer;
- ▶ every fixed $d \geq 2$: de Sitter volume–trace lemma, fixed F2 constant and constant-margin F3 failure;
- ▶ exact $V_3(\tau)$ and $V_4(\tau)$ formulas checked symbolically;
- ▶ both standard curvature-norm readings pass;
- ▶ O3: explicit regional reading retained;
- ▶ O5: proved modulo that O3 scope.

No simulation, fitted threshold or asymptotic-rate-only claim is used.

What we ask you to adjudicate

- 1. Admissibility in the original causal-set sense.** Is an induced-order embedding with the stated number–volume control still “faithful” or “Planck-scale uniform” when $\Theta(\rho^{1/d})$ deterministic points are aligned along one timelike geodesic?
- 2. Foundational prior art.** Do you know an earlier construction with the same configuration-level certificate: exact order + quantitative count uniformity + an explicit longest-chain/proper-time failure?

Separate author query. The exact regional scope intended by Madsen’s Remark 5.4 is being put to Madsen, not to you.

Closest results we found

Müller

Already has the dangerous conjunction: his Theorem 2 mechanism leaves the d^- Hauptvermutung false even under “Planck-scale uniformness”, a count condition on causally convex subsets, while proper-time geometry can be macroscopically wrong. The exact separation is different: fixed K , two geometries and nearby order laws; no fixed embedding with exact induced order and a chain-height/F3 certificate. His mechanism is a conformal deformation near a maximizer, not $\Theta(\rho^{1/d})$ planted points.

Henson

Changing a small number of causal relations can destroy embeddability; not the addition of aligned elements.

Braun

“Order + number = geometry” from laws of every sample size, under $d \geq 3$; different quantifiers and object.

Aghili et al.

Chain distributions for uniform sprinklings in several dimensions; no adversarial F2-without-F3 construction.

MÜLLER: CLOSEST NEGATIVE PRECURSOR

DOES NOT SUBSUME THIS CERTIFICATE

A reply can take one of three precise forms

- A.** Yes: under the stated reading, this is a genuine F1+F2-not-F3 counterexample in the original causal-set sense.
- B.** No: faithful or Planck-scale-uniform embedding intends an exclusion; please identify it.
- C.** Known: an earlier source already contains this same mechanism; please identify it.

Two source-level qualifications remain explicit:

CURVATURE_NORM_UNDEFINED / BOTH_STANDARD_READINGS_PASS

REGIONAL_SCOPE_NOT_LITERAL / EXPLICIT_REGIONAL_READING_ADOPTED

Claim discipline

No absolute priority claim is made. The mathematical claim is conditional on the two readings printed above. The requested expert judgment concerns interpretation and precedent, not an appeal to authority in place of proof.